

Writing down an entanglement in bra-ket notation

Asked 10 years, 9 months ago Modified 10 years, 9 months ago Viewed 2k times

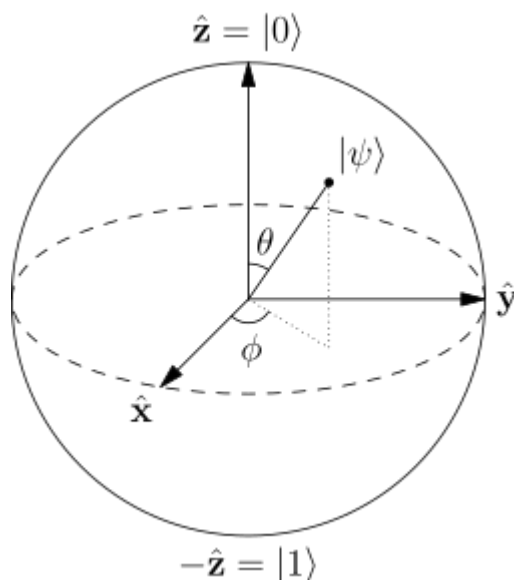


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I have a relatively complex (for me anyhow) situation which I would like to capture in bra-ket notation, rather than in words. However, I've not been able to find a source which will help me understand better how to do so, and I was hoping that one of you might help me out. I'll first describe exactly what I want to express, so that it's a little more clear what I mean. Note that this is not a homework question or something, it has to do with an experiment that I came up with, so chances are that it might be somewhat impossible to actually write it out the way I want to.

I have a system that lives on the [Bloch sphere](#), like this one, but with $|\uparrow\rangle$ instead of $|0\rangle$ and $|\downarrow\rangle$ instead of $|1\rangle$



Now, initially I am able to fully initialize something, say a spin, into 1 state, being $|0\rangle$. Then, I rotate the state by $\pi/2$ around the x-axis, and I wait a time t . In this time t , some more complex stuff happens. Depending on the state of a second spin, stuff will happen. This second spin can be in either of three states, $|-1\rangle$, $|0\rangle$ and $|1\rangle$.

Now, during this time t , the state of the first spin will precess about the Bloch sphere depending on the state of the second spin. If the second spin is in state $|-1\rangle$ the first state will rotate around the z-axis clock wise, if the second spin is in state $|0\rangle$ the first spin will not change, and if the second spin is in state $|1\rangle$ the first spin will rotate about the z-axis counterclockwise.

So, after this time t , I will rotate the first spin again, but this time by $\pi/2$ around the y-axis. This means that if I choose t such that it is exactly a quarter of the period in which the spin

makes a full circle, the two spin states will be entangled. I can use this to establish what the second spin state is by reading out the first spin state:

If $|-1\rangle$, then I will readout $|\uparrow\rangle$ with $P = 0$, if $|0\rangle$, then I will readout $|\uparrow\rangle$ with $P = 0.5$, and if $|1\rangle$, then I will readout $|\uparrow\rangle$ with $P = 1$.

I understand that this probably might be a little confusing, which is of course also the primary reason why I would just want to write this in nice, clean bra-ket notation. If there's anything in particular that's not clear, please let me know. And if someone could help me get started (possibly by pointing me to a similar example) I'd be very grateful.

Edit: Alright, I've done a little bit of reading on what I could find, and this is how far I get now

$$\text{Initially: } |\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = |\uparrow\rangle \otimes \frac{1}{\sqrt{3}}(|-1\rangle + |0\rangle + |1\rangle)$$

Rotate first spin by $\pi/2$ around x-axis

$$|\psi\rangle = R_x\left(\frac{\pi}{2}\right) |\psi_1\rangle \otimes |\psi_2\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - i|\downarrow\rangle) \otimes \frac{1}{\sqrt{3}}(|-1\rangle + |0\rangle + |1\rangle)$$

But here I get stuck again, as it is here that the rotation conditional on the second spin has to happen, and that I don't know how to do.

quantum-mechanics

quantum-information

quantum-spin

density-operator

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edited Mar 10, 2017 at 9:42



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1

asked Feb 23, 2014 at 21:25



user129412

1,581 1 14 29

Of course it would be nicest if I were able to do this for a general rotation angle ϕ and a general time t (a general axis will make it rather complicated, either x or y is fine for my purposes) – user129412 Feb 23, 2014 at 22:22

2 Answers

Sorted by: Highest score (default)



5

Okay, I don't quite get the details of what you are doing, but since this is linear algebra, I'd advise you to use linear algebra. You can then easily transfer between bra-ket notation and matrices.



First, let's fix what we are talking about: You have one system A containing one spin, so the system is a space $\mathbb{C}^2$ with basis states $|0\rangle, |1\rangle$ (down and up - you can of course name them down and up, but that doesn't change anything). Furthermore, you have a second system B with a state that can be in either $|-1\rangle, |0\rangle, |1\rangle$, i.e. a $\mathbb{C}^3$.



This means, your states live in $\mathbb{C}^2 \otimes \mathbb{C}^3$ and your time evolutions will just be some unitaries of this space. In order to write them down more easily, let's choose an ordering of the basis

$|0\rangle \otimes |-1\rangle, |0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |-1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle$, which means that $|0\rangle \otimes |-1\rangle$ corresponds to the vector $(1, 0, 0, 0, 0, 0)^{\text{tr}} \in \mathbb{C}^2 \otimes \mathbb{C}^3$.

Choosing such a basis makes it easier, to write down the corresponding unitaries. Let me make two examples:

You rotate the first spin $\pi/2$ around the x-axis. A rotation of a spin around the x-axis is nothing but the unitary evolution of the Pauli-x-axis. You can show that a rotation of the Bloch-sphere around an axis $\mathbf{n}$ by an angle θ is given by

$$R_{\mathbf{n}}(\theta) = e^{-i\theta\boldsymbol{\sigma}\cdot\mathbf{n}/2}$$

where $\boldsymbol{\sigma}$ is the vector of Pauli-matrices.

Thus, a simple rotation of the first part of the system by $\pi/2$ around the x-axis is given by the unitary

$$e^{-i\pi X/4} \otimes 1_3 \in \mathcal{B}(\mathbb{C}^2 \otimes \mathbb{C}^3)$$

with the identity acting on the second system.

Now, suppose you want to implement a conditional unitary. Well, nothing easier than that, you just make it up. You know where your basis states should end up (you just write down what the state will look like after the application of the conditional unitary for any of the basis states $|0\rangle \otimes |-1\rangle, |0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |-1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle$. This will give you all the entries of a 6×6 unitary corresponding to the operator. Since you have a parameter t , your unitary will be t -dependent.

This lets you create unitaries for every step of the way. Now to get the overall unitary of the whole process, you just have to multiply them all from right to left (later processes are multiplied from the left) - like a general quantum circuit.

In principle, now, in order to get your final state, you just multiply the matrix and your initial state vector. There might be a caveat here - I'm not entirely sure, whether you can actually initialize the whole system (i.e. both parts of the system are in some specific state - maybe a superposition, maybe not). If you can, then this will correspond to some vector, if you can't you'll need to use density matrices.

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answered Feb 24, 2014 at 13:53



Martin

15.8k

6

55

75

Thank you for your great answer. I'll try and work it out this way, see if I can pull it off. It sounds like this is indeed the way to go. – [user129412](#) Feb 24, 2014 at 13:59

I understand most of what you've written, but I get a little confused at the part where you describe how to make up the conditional. I just realized that what I wanted to write here is too long, so I've edited it onto the bottom of my 'answer' following yours. – [user129412](#) Feb 24, 2014 at 15:13



Following what I had been explained above, I think I might be able to solve my issue by writing it as such:

1



Initially:

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = |\uparrow\rangle \otimes \frac{1}{\sqrt{3}}(|-1\rangle + |0\rangle + |1\rangle)$$



Rotating the first spin by $\pi/2$ around x-axis

$$|\psi\rangle = \left(e^{-i\frac{\pi}{4}X} \otimes I\right) \otimes \left(|\uparrow\rangle \otimes \frac{1}{\sqrt{3}}(|-1\rangle + |0\rangle + |1\rangle)\right) = \frac{1}{\sqrt{6}}(|\uparrow\rangle - i|\downarrow\rangle) \otimes (|-1\rangle + |0\rangle + |1\rangle)$$

Conditional rotation:

$$\begin{aligned} |\psi\rangle &= R_{con} \frac{1}{\sqrt{6}}(|\uparrow\rangle - i|\downarrow\rangle) \otimes (|-1\rangle + |0\rangle + |1\rangle) \\ &= \frac{1}{\sqrt{6}}[(|\uparrow\rangle + |\downarrow\rangle) \otimes |-1\rangle + (|\uparrow\rangle - i|\downarrow\rangle) \otimes |0\rangle + (|\uparrow\rangle - |\downarrow\rangle) \otimes |1\rangle] \end{aligned}$$

Rotating the first spin by $\pi/2$ around y-axis

$$\begin{aligned} |\psi\rangle &= \left(e^{-i\frac{\pi}{4}Y} \otimes I\right) \otimes \frac{1}{\sqrt{6}}[(|\uparrow\rangle + |\downarrow\rangle) \otimes |-1\rangle + (|\uparrow\rangle - i|\downarrow\rangle) \otimes |0\rangle + (|\uparrow\rangle - |\downarrow\rangle) \otimes |1\rangle] \\ &= \frac{1}{\sqrt{3}}|\downarrow\rangle \otimes |-1\rangle + \frac{1}{\sqrt{3}}|\uparrow\rangle \otimes |1\rangle + \frac{1}{\sqrt{6}}(|\uparrow\rangle + |\downarrow\rangle) \otimes |0\rangle \end{aligned}$$

Here, I have yet to figure out what the actual shape of the conditional rotation is, so that's still left. Could anyone see whether or not this makes sense, what I've written down?

For the conditional, I know that what I want to do is such that $R(t)$ works like indicated below:

$$R_{con}(1/4) (|\uparrow\rangle - i|\downarrow\rangle) \otimes |-1\rangle = (|\uparrow\rangle + |\downarrow\rangle) \otimes |-1\rangle$$

$$R_{con}(1/4) (|\uparrow\rangle - i|\downarrow\rangle) \otimes |0\rangle = (|\uparrow\rangle - i|\downarrow\rangle) \otimes |0\rangle$$

$$R_{con}(1/4) (|\uparrow\rangle - i|\downarrow\rangle) \otimes |1\rangle = (|\uparrow\rangle - |\downarrow\rangle) \otimes |1\rangle$$

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edited Feb 24, 2014 at 15:18

answered Feb 24, 2014 at 14:53

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user129412

1,581 1 14 29

Should I literally just write out the entire 6x6 matrix that R_{con} is, and solve for all its entries?

– user129412 Feb 24, 2014 at 15:29

that is actually one way to do it - although you can more or less read off the entries. Another way to do it is to note that $R_{con -1}(t) \otimes |-1\rangle\langle -1| + R_{con 0}(t) \otimes |0\rangle\langle 0| + R_{con 1}(t) \otimes |1\rangle\langle 1|$ with the 2x2 conditional unitaries R_{con} does the trick (it's unitary, since your basis is orthogonal). That just leaves figuring out what unitaries act conditioned to what. – Martin Feb 25, 2014 at 9:32

Thanks a lot for your help Martin, it has been very valuable. – [user129412](#) Feb 25, 2014 at 11:15
